

Technical SEO Field Guide

Sitemaps of dead URLs, blocked noindex, wrong canonicals and soft 404s.

4 fixes, each one a written note with diagrams and a small Python and Node.js script that finds the problem and repairs it. All 4 have a script in the public repo.

Before you run anything. Every script starts in a dry run: it reports what it would change and writes nothing. Read that output first, then set `DRY_RUN=false` when you are satisfied it has understood your data.

Scripts: github.com/allanninal/technical-seo-fixes

Guides: allanninal.dev/seo

All 14 repos: github.com/allanninal

The fixes

1 a canonical tag pointing at staging, a redirect or nothing

Search Console says Alternate page with proper canonical tag for pages you very much want indexed, or your live pages are nominating a staging host nobody can reach. Canonical tags are almost always generated from a single base URL, so when that base is wrong the whole site is wrong at once — and the page looks completely normal in a browser, because a canonical is invisible.

[canonical_audit.py](#)

<https://www.allanninal.dev/seo/canonical-points-at-staging-or-a-redirect/>

<https://github.com/allanninal/technical-seo-fixes/tree/main/canonical-points-at-staging-or-a-redirect>

2 robots.txt blocks the noindex it was meant to enforce

You added noindex to a page and blocked it in robots.txt for good measure. Weeks later it is still in the results, listed without a description, and Search Console says Indexed, though blocked by robots.txt. The two rules cancelled each other out. Disallow prevents the crawl, so the crawler never fetches the page and never sees the noindex. Belt and braces removed the belt.

[robots_conflict.py](#)

<https://www.allanninal.dev/seo/robots-txt-blocks-the-noindex-you-added/>

<https://github.com/allanninal/technical-seo-fixes/tree/main/robots-txt-blocks-the-noindex-you-added>

3 **your sitemap lists URLs that redirect, 404 or say noindex**

Search Console is reporting Submitted URL marked 'noindex', or Submitted URL not found (404), and the count keeps growing. A sitemap is a statement that these URLs are worth indexing. Listing a URL that returns 404, redirects elsewhere, or carries a noindex is a contradiction — you are recommending a page and simultaneously telling the crawler to ignore it.

[sitemap_prune.py](#)

<https://www.allanninal.dev/seo/sitemap-lists-urls-that-must-not-be-indexed/>

<https://github.com/allanninal/technical-seo-fixes/tree/main/sitemap-lists-urls-that-must-not-be-indexed>

4 **a missing page that returns 200 instead of 404**

Search Console is reporting Soft 404 on pages that look fine, or your index count is far higher than the number of pages you actually have. Somewhere a catch-all is answering every request with 200 OK and a page that says “not found” in the body. To a crawler that is a real page with thin content, so it gets crawled, judged and sometimes indexed.

[soft_404_probe.py](#)

<https://www.allanninal.dev/seo/soft-404-returns-200/>

<https://github.com/allanninal/technical-seo-fixes/tree/main/soft-404-returns-200>
